

Worksheet No. - 02

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Aim/Overview of the practical: To implement the Depth-First Search (DFS) algorithm to traverse a given graph starting from a specified node, explore all reachable vertices by going as deep as possible along each branch before backtracking, and understand its working principle using stack/recursion for systematic graph traversal.

Objective: To design and implement the Depth-First Search (DFS) algorithm for graph traversal, starting from a given source node, in order to systematically visit all reachable vertices using recursion or a stack and to analyze the traversal process for understanding graph exploration techniques.

Procedure/Algorithm/Code

Algorithm:

1. Start the program.
2. Create a graph using an adjacency list or adjacency matrix.
3. Initialize an empty set or array to keep track of visited nodes.
4. Choose a starting node for DFS traversal.
5. Visit the starting node and mark it as visited.
6. For each unvisited adjacent node of the current node, do the following:
 - Recursively apply the DFS algorithm to that node.
7. Continue the process until all reachable nodes from the starting node are visited.
8. Backtrack when a node has no unvisited adjacent nodes.
9. Stop the algorithm after all nodes are traversed.

Code:

```
import matplotlib.pyplot as plt

import networkx as nx

graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F'],
    'D': [],
    'E': ['F'],
    'F': []
}

visited = set()

def dfs(node):
    if node not in visited:
        print(node, end=' ')
        visited.add(node)
        for neighbor in graph[node]:
            dfs(neighbor)

start_node = 'A'

print("DFS Traversal starting from node", start_node, ":")

dfs(start_node)

G = nx.DiGraph()

for node in graph:
    for neighbor in graph[node]:
```

```
G.add_edge(node, neighbor)
```

```
plt.figure(figsize=(8, 5))
```

```
pos = nx.spring_layout(G)
```

```
nx.draw(G, pos, with_labels=True, node_color='lightblue',
```

```
node_size=1500, edge_color='gray',
```

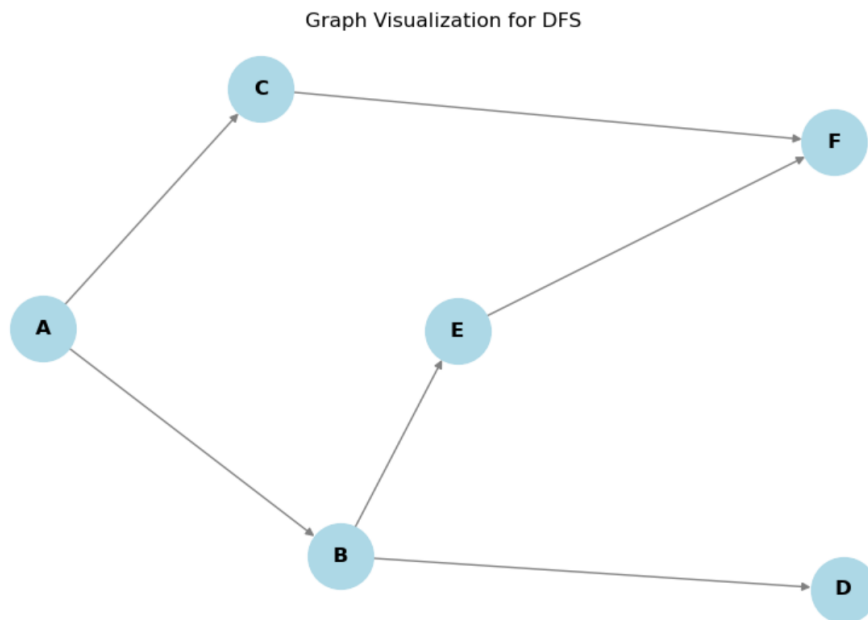
```
font_size=12, font_weight='bold')
```

```
plt.title("Graph Visualization for DFS")
```

```
plt.show()
```

Output

DFS Traversal starting from node A :
A B D E F C



Learning outcomes (What I have learnt):

1. Understood the concept of Depth-First Search (DFS) and its role in graph traversal.
2. Learnt how DFS explores a graph by going deep into each branch before backtracking.
3. Gained practical experience in implementing DFS using recursion or a stack.